

Programming square wave & sawtooth wave in R

Asked 9 years, 5 months ago Modified 6 years ago Viewed 3k times  Part of R Language Collective

I'm trying to build a function in R that represents square and sawtooth waves with 100Hz in range [0,1]. I tried this:

```
1 squarewave <- function (t) {
  # 0.01 == 100Hz(=2Pi) -> 1 Period of the Squarewave
  # 0.005 == Pi -> Half Period of the Squarewave
  # if t smaller than a half period -> 1
  # if t greater or equal than half a period -> 0
  if ((t %% 0.01) < 0.005)
    return (1)
  else if ((t %% 0.01) >= 0.005)
    return (0)
}
```

When I try to plot this function with:

```
plot(squarewave)
```

I get the following error:

```
> plot(squarewave)
Error in curve(expr = x, from = from, to = to, xlim = xlim, ylab = ylab, :
'expr' has not been evaluated to an object of length 'n'
In addition: Warning message:
In if ((t%%0.01) < 0.005) return(1) else if ((t%%0.01) >= 0.005) return(0) :
the condition has length > 1 and only the first element will be used
```

So why does this not work?

 **waveform** [Edit tags](#)

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edited Oct 12, 2018 at 6:42

asked May 7, 2015 at 14:34



Cœur
38.4k 26 201 275



tbol
145 1 11

 Post your error in english, not in german.. – [Naruto](#) May 7, 2015 at 14:41

1  You aren't passing any parameters to `squarewave` when you plot it. You can't plot a function directly like that. – [Alex A.](#) May 7, 2015 at 14:44

2 Answers

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 I was wrong initially, wanted to update.

0

Unless it is properly vectorized, you cannot plot a raw function and need to instead plot the output of the function. A simple way of doing this with your specific function is below.

```
sequence <- seq(from = 0, to = 0.01, by = 0.00001)
plot(sapply(X = sequence, FUN = squarewave), type = "o")
```

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edited May 7, 2015 at 15:03

answered May 7, 2015 at 14:45



TAREhman
6,719 3 35 60

 that's not right: `cos64hz <- function (t) { return (cos(2*64*pi*t)) }` works great with: `plot(cos64hz, type="l", xlim=c(0,1/20), ylim=c(-1,1.5), col="blue", ylab = "Amplitude", xlab="Zeit [sec]")` – [tbol](#) May 7, 2015 at 14:48 

 Could you be more specific? I produced a very nice square wave using it. – [TAREhman](#) May 7, 2015 at 14:49

 @AlexA. The function as written won't accept a vector as an argument, hence the use of `sapply()` to get a vector. – [TAREhman](#) May 7, 2015 at 14:54

 @TAREhman: Oh yeah you're right. The internals should use `ifelse`, then a straight call to `plot` would be ideal. – [Alex A.](#) May 7, 2015 at 14:58

1  @tbol Can you edit your comment into your question? It's relevant there for anyone who is trying to solve this. EDITED TO ADD: My guess is that the function you posted in your question is vectorized and R is able to interpret it; the other one is not and fails. – [TAREhman](#) May 7, 2015 at 14:59 

|

 You need to pass a vectorised function to `plot`. So either use `Vectorize` to do this automatically, or use `ifelse` instead of `if`.

2

```
plot(Vectorize(squarewave))
```



or



```
squarewave2 <- function (t) {  
  # 0.01 == 100Hz(=2Pi) -> 1 Period of the Squarewave  
  # 0.005 == Pi -> Half Period of the Squarewave  
  # if t smaller than a half period -> 1  
  # if t greater or equal than half a period -> 0  
  
  ifelse(((t %% 0.01) < 0.005),1,0)  
}  
  
plot(squarewave2)
```

To increase the resolution of the `plot`, use the parameter `n`, see `?curve` for details.

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answered May 7, 2015 at 14:51



James

66.6k

14

157

196



Sorry, I figured it out and deleted my comment before I noticed you had answered. Anyway, I turned up `n` as you suggested and it worked fine. I did not know this before so I learned something today from your answer. – [TARehman](#) May 7, 2015 at 15:07



That's the point! When using a `ifelse`-Statement the function appears as a vectorized function that can be plotted. Next, the plot seemed like a trapeze. To let the plot look like a real square wave you have to increase the `n`-Value of function `plot`. Thanks to James! – [tbol](#) May 7, 2015 at 16:12